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Wearable device with light source and optical sensor

Abstract

A wearable device with a light source and an optical sensor. In some embodiments, a system includes a wearable device for being worn by a subject and for measuring a first biomarker of a blood vessel of the subject. The wearable device may include a first light source, and a plurality of optical detectors. The system may be configured: to perform a photoplethysmography measurement with each of the plurality of optical detectors, the first biomarker being a blood volume; or to perform a speckleplethysmography measurement with each of the plurality of optical detectors, the first biomarker being a blood flow velocity.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4674011	12/1986	Patton et al.	N/A	N/A
5243983	12/1992	Tarr et al.	N/A	N/A
5497769	12/1995	Gratton et al.	N/A	N/A
5532860	12/1995	Hershey et al.	N/A	N/A
5772587	12/1997	Gratton et al.	N/A	N/A
5830132	12/1997	Robinson	N/A	N/A
6154259	12/1999	Hargis et al.	N/A	N/A
6243601	12/2000	Wist	N/A	N/A
6246892	12/2000	Chance	N/A	N/A
6256016	12/2000	Piot et al.	N/A	N/A
6560478	12/2002	Alfano et al.	N/A	N/A
6919549	12/2004	Bamji et al.	N/A	N/A
7035679	12/2005	Addison et al.	N/A	N/A
7113817	12/2005	Winchester, Jr. et al.	N/A	N/A
7202466	12/2006	Babayoff et al.	N/A	N/A
7250317	12/2006	Heideman	N/A	N/A
7295783	12/2006	Singh et al.	N/A	N/A
7375812	12/2007	Atia et al.	N/A	N/A
7474407	12/2008	Gutin	N/A	N/A
7505128	12/2008	Zribi et al.	N/A	N/A
7616984	12/2008	Barbour et al.	N/A	N/A
7761126	12/2009	Gardner et al.	N/A	N/A
7865225	12/2010	Kaltschmidt et al.	N/A	N/A
7922664	12/2010	Elliott	N/A	N/A
7925056	12/2010	Presura et al.	N/A	N/A
8237927	12/2011	Reeve et al.	N/A	N/A
8277384	12/2011	Fine	N/A	N/A
8313439	12/2011	McCombie et al.	N/A	N/A
8343062	12/2012	Fortin et al.	N/A	N/A
8343063	12/2012	Borgos	N/A	N/A
8376955	12/2012	Baker, Jr.	N/A	N/A
8398556	12/2012	Sethi et al.	N/A	N/A
8868149	12/2013	Eisen et al.	N/A	N/A
8920332	12/2013	Hong et al.	N/A	N/A
8923942	12/2013	Bernreuter	N/A	N/A
8945017	12/2014	Venkatraman et al.	N/A	N/A

8948832	12/2014	Hong et al.	N/A	N/A
8956303	12/2014	Hong et al.	N/A	N/A
8998815	12/2014	Venkatraman et al.	N/A	N/A
9005129	12/2014	Venkatraman et al.	N/A	N/A
9113794	12/2014	Hong et al.	N/A	N/A
9113795	12/2014	Hong et al.	N/A	N/A
9149216	12/2014	Eisen et al.	N/A	N/A
9155480	12/2014	Thakor et al.	N/A	N/A
9226673	12/2015	Ferguson, Jr. et al.	N/A	N/A
9237855	12/2015	Hong et al.	N/A	N/A
9307917	12/2015	Hong et al.	N/A	N/A
9494567	12/2015	Islam	N/A	N/A
9687162	12/2016	Vetter et al.	N/A	N/A
9704050	12/2016	Lee et al.	N/A	N/A
9730622	12/2016	Eisen et al.	N/A	N/A
9772280	12/2016	Cerussi et al.	N/A	N/A
9804027	12/2016	Fish et al.	N/A	N/A
9846126	12/2016	Gunn, III et al.	N/A	N/A
9848787	12/2016	White et al.	N/A	N/A
9851298	12/2016	Isikman et al.	N/A	N/A
9877681	12/2017	Silverman	N/A	N/A
9931040	12/2017	Homyk et al.	N/A	N/A
9970955	12/2017	Homyk et al.	N/A	N/A
10004406	12/2017	Yuen et al.	N/A	N/A
10058256	12/2017	Chen et al.	N/A	N/A
10178959	12/2018	Homyk et al.	N/A	N/A
10178973	12/2018	Venkatraman et al.	N/A	N/A
10194808	12/2018	Thompson et al.	N/A	N/A
10206576	12/2018	Shcherbakov et al.	N/A	N/A
10215698	12/2018	Han et al.	N/A	N/A
10241033	12/2018	Uematsu et al.	N/A	N/A
10271740	12/2018	Ward et al.	N/A	N/A
10314532	12/2018	Ward et al.	N/A	N/A
10326035	12/2018	Lu et al.	N/A	N/A
10326036	12/2018	Sweeney et al.	N/A	N/A
10349847	12/2018	Kwon et al.	N/A	N/A
10352768	12/2018	Simpkin et al.	N/A	N/A
10357165	12/2018	Yoon	N/A	N/A
10420498	12/2018	Horstmeyer et al.	N/A	N/A
10422693	12/2018	Fish et al.	N/A	N/A
10451537	12/2018	Nakaji	N/A	N/A
10463286	12/2018	Schenkman et al.	N/A	N/A
10492684	12/2018	Khachaturian et al.	N/A	N/A
10506926	12/2018	Khachaturian et al.	N/A	N/A
10506955	12/2018	Tholl et al.	N/A	N/A
10568527	12/2019	Yoon et al.	N/A	N/A
10588519	12/2019	Yuen et al.	N/A	N/A
10602987	12/2019	Khachaturian et al.	N/A	N/A
10627849	12/2019	Scofield et al.	N/A	N/A
10641962	12/2019	Nykänen et al.	N/A	N/A

10643903	12/2019	Drake et al.	N/A	N/A
10667688	12/2019	Khachaturian et al.	N/A	N/A
10677989	12/2019	Abediasl et al.	N/A	N/A
10681259	12/2019	Ichiki et al.	N/A	N/A
10681283	12/2019	Nakashima et al.	N/A	N/A
10694997	12/2019	Kim et al.	N/A	N/A
10718668	12/2019	Gu et al.	N/A	N/A
10722177	12/2019	Homyk et al.	N/A	N/A
10739256	12/2019	Rickman et al.	N/A	N/A
10750956	12/2019	Zalevsky et al.	N/A	N/A
10775239	12/2019	Lee et al.	N/A	N/A
10813597	12/2019	Rice et al.	N/A	N/A
10820858	12/2019	Yoon et al.	N/A	N/A
10842422	12/2019	Yu et al.	N/A	N/A
10871503	12/2019	Homyk et al.	N/A	N/A
10895525	12/2020	Swanson	N/A	N/A
10966616	12/2020	De Morree et al.	N/A	N/A
10973422	12/2020	Pantelopoulos et al.	N/A	N/A
11022751	12/2020	Bauters et al.	N/A	N/A
11045103	12/2020	Shchekin et al.	N/A	N/A
11079364	12/2020	Leger et al.	N/A	N/A
11096601	12/2020	Hong et al.	N/A	N/A
11096608	12/2020	Van Dorpe et al.	N/A	N/A
11129544	12/2020	Zalevsky et al.	N/A	N/A
11202582	12/2020	Verkruijse et al.	N/A	N/A
11213217	12/2021	Han et al.	N/A	N/A
11278220	12/2021	Tucker et al.	N/A	N/A
11298035	12/2021	Huijbregts et al.	N/A	N/A
11369275	12/2021	Song et al.	N/A	N/A
11445922	12/2021	Naima	N/A	N/A
11553851	12/2022	Kim et al.	N/A	N/A
11583185	12/2022	Homyk et al.	N/A	N/A
11666238	12/2022	Rege et al.	N/A	N/A
11666277	12/2022	Yoon et al.	N/A	N/A
11684281	12/2022	Pantelopoulos et al.	N/A	N/A
11690513	12/2022	Hu et al.	N/A	N/A
11696693	12/2022	Wong	N/A	N/A
11709120	12/2022	Rice et al.	N/A	N/A
11744491	12/2022	Dunn et al.	N/A	N/A
11751811	12/2022	Sun et al.	N/A	N/A
11759116	12/2022	White et al.	N/A	N/A
11759121	12/2022	Mccann et al.	N/A	N/A
11771343	12/2022	Sacha	N/A	N/A
11800990	12/2022	White et al.	N/A	N/A
11857301	12/2023	Homyk et al.	N/A	N/A
11883134	12/2023	Leabman	N/A	N/A
11890081	12/2023	Jang	N/A	N/A
11980451	12/2023	Albert	N/A	N/A
12109006	12/2023	Dunn et al.	N/A	N/A
2002/0195496	12/2001	Tsikos et al.	N/A	N/A

2003/0052169	12/2002	Tsikos et al.	N/A	N/A
2003/0137669	12/2002	Rollins et al.	N/A	N/A
2005/0249509	12/2004	Nagarajan et al.	N/A	N/A
2006/0124829	12/2005	Song et al.	N/A	N/A
2006/0132790	12/2005	Gutin	N/A	N/A
2006/0204175	12/2005	Laurent-Lund et al.	N/A	N/A
2006/0247514	12/2005	Panasyuk et al.	N/A	N/A
2007/0051601	12/2006	Wang et al.	N/A	N/A
2007/0057182	12/2006	Feuerbaum	N/A	N/A
2007/0093702	12/2006	Yu et al.	N/A	N/A
2008/0097172	12/2007	Sawada et al.	N/A	N/A
2008/0154126	12/2007	Culver et al.	N/A	N/A
2008/0204752	12/2007	Dorvee et al.	N/A	N/A
2008/0220512	12/2007	Koh et al.	N/A	N/A
2008/0316567	12/2007	Grasser et al.	N/A	N/A
2009/0177094	12/2008	Brown et al.	N/A	N/A
2009/0202251	12/2008	Shibayama	N/A	N/A
2009/0209834	12/2008	Fine	N/A	N/A
2009/0284748	12/2008	Melman et al.	N/A	N/A
2010/0004741	12/2009	Gupta et al.	N/A	N/A
2010/0046234	12/2009	Abu-Ageel	N/A	N/A
2010/0056928	12/2009	Zuzak et al.	N/A	N/A
2010/0226646	12/2009	Chan et al.	N/A	N/A
2011/0054277	12/2010	Pinter et al.	N/A	N/A
2011/0082355	12/2010	Eisen et al.	N/A	N/A
2011/0087108	12/2010	Onoe et al.	N/A	N/A
2011/0196244	12/2010	Ribas Ripoll et al.	N/A	N/A
2012/0130215	12/2011	Fine et al.	N/A	N/A
2012/0232402	12/2011	MacFarlane et al.	N/A	N/A
2013/0131475	12/2012	Eisen et al.	N/A	N/A
2013/0190630	12/2012	Borgos	N/A	N/A
2013/0204112	12/2012	White et al.	N/A	N/A
2013/0278631	12/2012	Border et al.	N/A	N/A
2014/0094666	12/2013	Fine	N/A	N/A
2014/0118695	12/2013	Shimada et al.	N/A	N/A
2014/0120319	12/2013	Joseph	N/A	N/A
2014/0200423	12/2013	Eisen et al.	N/A	N/A
2014/0313524	12/2013	Banyay et al.	N/A	N/A
2014/0316286	12/2013	Addison et al.	N/A	N/A
2014/0376001	12/2013	Swanson	N/A	N/A
2015/0157224	12/2014	Carmon et al.	N/A	N/A
2015/0196251	12/2014	Outwater et al.	N/A	N/A
2015/0201854	12/2014	Hong et al.	N/A	N/A
2015/0323311	12/2014	Muijs et al.	N/A	N/A
2016/0058300	12/2015	Yoon et al.	N/A	N/A
2016/0066790	12/2015	Shcherbakov et al.	N/A	N/A
2016/0106327	12/2015	Yoon et al.	N/A	N/A
2016/0157736	12/2015	Huang et al.	N/A	N/A
2016/0161685	12/2015	Xu et al.	N/A	N/A
2016/0183882	12/2015	Henley et al.	N/A	N/A

2016/0195473	12/2015	Fujiwara et al.	N/A	N/A
2016/0242647	12/2015	Ishii et al.	N/A	N/A
2016/0266337	12/2015	Feng	N/A	N/A
2016/0278676	12/2015	Eisen et al.	N/A	N/A
2016/0282265	12/2015	Su et al.	N/A	N/A
2016/0287107	12/2015	Szabados et al.	N/A	N/A
2016/0360966	12/2015	Ishii et al.	N/A	N/A
2017/0007138	12/2016	Kim et al.	N/A	N/A
2017/0014037	12/2016	Coppola et al.	N/A	N/A
2017/0065184	12/2016	Barak	N/A	N/A
2017/0105618	12/2016	Schmoll et al.	N/A	N/A
2017/0108439	12/2016	Stievater et al.	N/A	N/A
2017/0138789	12/2016	Ivanov	N/A	N/A
2017/0164878	12/2016	Connor	N/A	N/A
2017/0188851	12/2016	LeBoeuf et al.	N/A	N/A
2017/0231513	12/2016	Presura et al.	N/A	N/A
2017/0315292	12/2016	Mullen et al.	N/A	N/A
2018/0020962	12/2017	Yu et al.	N/A	N/A
2018/0045566	12/2017	Fish et al.	N/A	N/A
2018/0110423	12/2017	Presura et al.	N/A	N/A
2018/0160913	12/2017	Fine	N/A	N/A
2018/0168465	12/2017	Yamada et al.	N/A	N/A
2018/0202927	12/2017	Isikman et al.	N/A	N/A
2018/0228363	12/2017	Friskén et al.	N/A	N/A
2018/0238794	12/2017	Kangas et al.	N/A	N/A
2018/0263519	12/2017	Gu	N/A	N/A
2018/0283950	12/2017	Ge et al.	N/A	N/A
2018/0296168	12/2017	Rice et al.	N/A	N/A
2019/0041736	12/2018	Grunnet-Jepsen et al.	N/A	N/A
2019/0046056	12/2018	Khachaturian et al.	N/A	N/A
2019/0053721	12/2018	Boas et al.	N/A	N/A
2019/0094009	12/2018	Aizawa et al.	N/A	N/A
2019/0094564	12/2018	Rivera et al.	N/A	N/A
2019/0167118	12/2018	Vilenskii et al.	N/A	N/A
2019/0175030	12/2018	Verkruijse et al.	N/A	N/A
2019/0336006	12/2018	Horstmeyer et al.	N/A	N/A
2019/0343442	12/2018	Aung et al.	N/A	N/A
2019/0343456	12/2018	Kahlert et al.	N/A	N/A
2019/0369650	12/2018	Swanson et al.	N/A	N/A
2019/0387972	12/2018	Hu et al.	N/A	N/A
2019/0391243	12/2018	Nicolaescu	N/A	N/A
2019/0391702	12/2018	Jo et al.	N/A	N/A
2020/0003619	12/2019	Hu et al.	N/A	N/A
2020/0011995	12/2019	Send et al.	N/A	N/A
2020/0069225	12/2019	Vizbaras et al.	N/A	N/A
2020/0100705	12/2019	Dellimore et al.	N/A	N/A
2020/0143534	12/2019	Wright et al.	N/A	N/A
2020/0158548	12/2019	Rice et al.	N/A	N/A
2020/0196874	12/2019	Rozental et al.	N/A	N/A
2020/0214602	12/2019	Narumi et al.	N/A	N/A

2020/0237272	12/2019	Lin et al.	N/A	N/A
2020/0249492	12/2019	Maes	N/A	N/A
2020/0323440	12/2019	Vule et al.	N/A	N/A
2020/0359948	12/2019	Dunn et al.	N/A	N/A
2020/0397351	12/2019	Miyata	N/A	N/A
2021/0000385	12/2020	Warren et al.	N/A	N/A
2021/0022623	12/2020	Rice et al.	N/A	N/A
2021/0028602	12/2020	Cao et al.	N/A	N/A
2021/0161408	12/2020	Wakita	N/A	N/A
2021/0186431	12/2020	Jung et al.	N/A	N/A
2021/0267471	12/2020	Bonomi et al.	N/A	N/A
2021/0321887	12/2020	Fukazawa et al.	N/A	N/A
2021/0330202	12/2020	Konecky	N/A	N/A
2021/0338083	12/2020	Sie et al.	N/A	N/A
2021/0386310	12/2020	Hong et al.	N/A	N/A
2021/0405518	12/2020	Lablans	N/A	N/A
2022/0015649	12/2021	Ikuta et al.	N/A	N/A
2022/0018762	12/2021	Ekin et al.	N/A	N/A
2022/0019861	12/2021	Durr et al.	N/A	N/A
2022/0039679	12/2021	Califa et al.	N/A	N/A
2022/0061644	12/2021	Fontaine et al.	N/A	N/A
2022/0104822	12/2021	Shelton, IV et al.	N/A	N/A
2022/0117557	12/2021	Hsu et al.	N/A	N/A
2022/0196557	12/2021	Zheng et al.	N/A	N/A
2022/0211286	12/2021	Tank et al.	N/A	N/A
2022/0265158	12/2021	Tokura	N/A	N/A
2022/0370010	12/2021	Zilkie et al.	N/A	N/A
2022/0413143	12/2021	Parsa et al.	N/A	N/A
2023/0003938	12/2022	Zilkie et al.	N/A	N/A
2023/0039055	12/2022	Gardner et al.	N/A	N/A
2023/0048766	12/2022	Frey	N/A	N/A
2023/0064006	12/2022	Kim et al.	N/A	N/A
2023/0087295	12/2022	Dunn et al.	N/A	N/A
2023/0148885	12/2022	Bechtel et al.	N/A	N/A
2023/0148886	12/2022	Bechtel et al.	N/A	N/A
2023/0164444	12/2022	Yang	N/A	N/A
2023/0225643	12/2022	Scofield et al.	N/A	N/A
2023/0277062	12/2022	Dalvi et al.	N/A	N/A
2023/0277075	12/2022	Pantelopoulos et al.	N/A	N/A
2023/0296510	12/2022	Xu	N/A	N/A
2023/0320598	12/2022	Khine et al.	N/A	N/A
2023/0347029	12/2022	Corso et al.	N/A	N/A
2023/0375525	12/2022	Merritt et al.	N/A	N/A
2023/0397818	12/2022	Newhouse et al.	N/A	N/A
2023/0401747	12/2022	Dunn et al.	N/A	N/A
2024/0032790	12/2023	Patel et al.	N/A	N/A
2024/0041342	12/2023	Lai et al.	N/A	N/A
2024/0074667	12/2023	Rick et al.	N/A	N/A
2024/0108289	12/2023	Bechtel et al.	N/A	N/A
2024/0115212	12/2023	Jang	N/A	N/A

2024/0156355	12/2023	O'Brien et al.	N/A	N/A
2024/0298907	12/2023	Bechtel et al.	N/A	N/A
2024/0350019	12/2023	Pery-Shechter et al.	N/A	N/A
2024/0364420	12/2023	Vallius et al.	N/A	N/A
2025/0025058	12/2024	Bechtel et al.	N/A	N/A
2025/0169696	12/2024	Dunn et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2 861 089	12/2020	CA	N/A
108709847	12/2017	CN	N/A
110301896	12/2018	CN	N/A
211094079	12/2019	CN	N/A
211131004	12/2019	CN	N/A
112639582	12/2020	CN	N/A
114466549	12/2021	CN	N/A
3 002 568	12/2015	EP	N/A
2 395 958	12/2016	EP	N/A
3 384 841	12/2017	EP	N/A
3 558 119	12/2019	EP	N/A
3 886 686	12/2020	EP	N/A
3 903 676	12/2020	EP	N/A
WO 2024/052289	12/2013	WO	N/A
WO 2018/029123	12/2017	WO	N/A
WO 2019/149815	12/2018	WO	N/A
WO 2019/233903	12/2018	WO	N/A
WO 2020/030641	12/2019	WO	N/A
WO 2020/114989	12/2019	WO	N/A
WO 2021/058338	12/2020	WO	N/A
WO 2021/094473	12/2020	WO	N/A
WO 2021/116766	12/2020	WO	N/A
WO 2021/116766	12/2020	WO	N/A
WO 2023/031927	12/2022	WO	N/A
WO 2023/245149	12/2022	WO	N/A
WO 2024/173585	12/2023	WO	N/A

OTHER PUBLICATIONS

European Patent Office Communication pursuant to Article 94(3) EPC, for Patent Application No. 22776935.3, dated Jun. 24, 2025, 7 pages. cited by applicant

U.S. Office Action from U.S. Appl. No. 18/991,054, dated Jun. 24, 2025, 23 pages. cited by applicant

Van Gastel, M. et al., "Camera-based pulse-oximetry—validated risks and opportunities from theoretical analysis", Biomedical Optics Express, Dec. 5, 2017, pp. 102-119, vol. 9, No. 1, Optical Society of America. cited by applicant

Website: "0.07mm Dia., TO-46 Package, InGaAs Photodiode", 2022, printed Dec. 7, 2022, 1 page, Edmund Optics Inc., <https://www.edmundoptics.com/p/ingaas-detector-70mum-dia-to-46/12571/>. cited by applicant

Website: "FlowMet Peripheral Blood Flow Monitoring System", updated Oct. 2022, printed Dec. 7, 2022, 7 pages, <https://www.medtronic.com/us-en/healthcare-professionals/products/cardiovascular/intraprocedural-monitoring/flowmet.html>. cited by applicant

Website: “Optical Solutions”, Molex, dated 2023, printed May 10, 2023, 13 pages, Molex, LLC, <https://www.molex.com/en-us/products/optical-solutions>. cited by applicant

Website: “Track Your SpO2 to Uncover Changes in Your Wellbeing”, Fitbit News, dated Sep. 7, 2020, printed Apr. 17, 2023, 7 pages, Fitbit, Inc., <https://blog.fitbit.com/track-your-spo2/>.626. cited by applicant

Wenz, J. J., “Examining water in model membranes by near infrared spectroscopy and multivariate analysis”, BBA—Biomembranes, Dec. 9, 2017, pp. 673-682, Elsevier B.V., <https://www.sciencedirect.com/science/article/pii/S0005273617303905>. cited by applicant

Xu, J. et al., “Interferometric speckle visibility spectroscopy (ISVS) for human cerebral blood flow monitoring”, APL Photonics, Dec. 4, 2020, pp. 126102-1 through 126102-10, vol. 5. AIP Publishing. cited by applicant

Xu, M. et al., “Laser Speckle Reduction Using a Motionless Despeckle Element Based on Random Mie Scattering”, Journal of Display Technology, Nov. 12, 2013, pp. 151-156, vol. 10, No. 2, IEEE. cited by applicant

Yamakoshi, Y. et al., “Side-scattered finger-photoplethysmography: experimental investigations toward practical noninvasive measurement of blood glucose”, Journal of Biomedical Optics, Jun. 2017, pp. 067001-1 through 067001-11, vol. 22, No. 6, SPIE. cited by applicant

Yao, Z. et al., “Integrated Silicon Photonic Microresonators: Emerging Technologies”, IEEE Journal of Selected Topics in Quantum Electronics, Jun. 11, 2018, 24 pages, vol. 24, No. 6, IEEE. cited by applicant

Zalevsky, Z. et al., “Novel Approaches for Near and Far Field Super Resolved Imaging”, 22nd Congress of the International Commission for Optics: Light for the Development of the World, Proc. of SPIE, Sep. 15, 2011, pp. 80116M-1 through 80116M-11, vol. 8011, No. 1, SPIE. cited by applicant

Zhang, J. et al., “III-V-on-Si photonic integrated circuits realized using micro-transfer-printing”, APL Photonics, Nov. 4, 2019, pp. 110803-1 through 110803-10. cited by applicant

Zijlstra, W. G. et al., “Absorption Spectra of Human Fetal and Adult Oxyhemoglobin, De-Oxyhemoglobin, Carboxyhemoglobin, and Methemoglobin”, Clinical Chemistry, Sep. 1991, pp. 1633-1638, vol. 37, No. 9, <https://academic.oup.com/clinchem/article-abstract/37/9/1633/5649610?redirectedFrom=fulltext>. cited by applicant

Zilkie, A. J. et al., “Multi-Micron Silicon Photonics Platform for Highly Manufacturable and Versatile Photonic Integrated Circuits”, IEEE Journal of Selected Topics in Quantum Electronics, Apr. 15, 2019, 13 pages, vol. 25, No. 5, IEEE. cited by applicant

Zilkie, A. J. et al., “Power-efficient III-V/Silicon external cavity DBR lasers”, Optics Express, Sep. 27, 2012, pp. 23456-23462, vol. 20, No. 21, Optical Society of America. cited by applicant

Akram, M. N. et al., “Laser speckle reduction due to spatial and angular diversity introduced by fast scanning micromirror”, Applied Optics, Jun. 4, 2010, pp. 3297-3304, vol. 49, No. 17, Optical Society of America. cited by applicant

Abookasis, D. et al., Feasibility study of hidden flow imaging based on laser speckle technique using multiperspectives contrast images, Optics and Lasers in Engineering, 2014, pp. 38-45. cited by applicant

Apsel, S. et al., “Rolling-Shutter Laser Speckle Analysis in Bio-Photonics”, Proc. of SPIE, Jun. 20, 2024, pp. 130060U-1-130060U-4, vol. 13006, SPIE. cited by applicant

Baek, H. J. et al., “The Effect of Optical Crosstalk on Accuracy of Reflectance-Type Pulse Oximeter for Mobile Healthcare”, Journal of Healthcare Engineering, Oct. 21, 2018, 9 pages, vol. 2018, Article ID 3521738, Hindawi, <https://doi.org/10.1155/2018/3521738>. cited by applicant

Baets, R. et al., “Spectroscopy-on-chip applications of silicon photonics”, Proc. Of SPIE, 2013, pp. 86270I-1 through 86270I-10, vol. 8627, SPIE. cited by applicant

Berger, A. J. et al., “Feasibility of measuring blood glucose concentration by near-infrared Raman spectroscopy”, Spectrochimica Acta Part A, 1997, pp. 287-292, Elsevier Science B.V. cited by

applicant

Bi, R. et al., "A speckle-based method for fast blood flow measurement in deep tissue", Proceedings of SPIE, Optical Biopsy XIX: Toward Real-Time Spectroscopic Imaging and Diagnosis, Mar. 5, 2021, pp. 1163606-1 through 1163606-5, vol. 11636, SPIE. cited by applicant

Bi, R. et al., "Fast pulsatile blood flow measurement in deep tissue through a multimode detection fiber", Journal of Biomedical Optics, May 13, 2020, pp. 055003-1 through 055003-10, vol. 25(5), SPIE. cited by applicant

Biswas, A. et al., "Fast diffuse correlation spectroscopy with a low-cost, fiber-less embedded diode laser", Biomedical Optics Express, Oct. 4, 2021, pp. 6686-6700, vol. 12, No. 11, Optical Society of America. cited by applicant

Brouckaert, J. et al., "Silicon-on-Insulator Microspectrometer", Proceedings Symposium IEEE/LEOS Benelux Chapter, 2008, pp. 7-10, IEEE. cited by applicant

Cole, D. B. et al., "Integrated heterodyne interferometer with on-chip modulators and detectors", Optics Letters, Jun. 25, 2015, pp. 3097-3100, vol. 40, No. 13, Optical Society of America. cited by applicant

Epping, J. P. et al., "High power, tunable, narrow linewidth dual gain hybrid laser", Laser Congress, Oct. 3, 2019, pp. 1-2. cited by applicant

European Patent Office Communication pursuant to Rule 114(2) EPC, for Patent Application No. 22776935.3, mailed Aug. 9, 2024, 6 pages. cited by applicant

Fu, D. et al., "In Vivo Metabolic Fingerprinting of Neutral Lipids with Hyperspectral Stimulated Raman Scattering Microscopy", Journal of the American Chemical Society, May 28, 2014, pp. 8820-8828, American Chemical Society Publications. cited by applicant

Fukui, T. et al., "Single-Pixel Imaging Using Multimode Fiber and Silicon Photonic Phased Array", Journal of Lightwave Technology, Jul. 14, 2020, pp. 839-844, vol. 39, No. 3, IEEE. cited by applicant

Ge, Z. et al., "Dynamic laser speckle analysis using the event sensor", Applied Optics, Dec. 23, 2020, pp. 172-178, vol. 60, No. 1, Optical Society of America. cited by applicant

Ghijsen, M. et al., "Wearable speckle plethysmography (SPG) for characterizing microvascular flow and resistance", Biomedical Optics Express, Jul. 30, 2018, pp. 3937-3952, vol. 9, No. 8, Optical Society of America under the terms of the OSA Open Access Publishing Agreement. cited by applicant

Goodman, J. W., "Some fundamental properties of speckle", Journal of the Optical Society of America, Nov. 1976, pp. 1145-1150, vol. 66, No. 11, Optical Society of America. cited by applicant

Gottschling, K. et al., "Molecular Insights into Carbon Dioxide Sorption in Hydrazone-Based Covalent Organic Frameworks with Tertiary Amine Moieties", Chemistry of Materials, Feb. 13, 2019, pp. 1946-1955, American Chemical Society. cited by applicant

Hashimoto, Y. et al., "Fabrication of an Anti-Reflective and Super-Hydrophobic Structure by Vacuum Ultraviolet Light-Assisted Bonding and Nanoscale Pattern Transfer", Micromachines, Apr. 15, 2018, pp. 1-11, www.mdpi.com/journal/micromachines. cited by applicant

Hollis, V. S. et al., "Non-invasive monitoring of brain tissue temperature by near-infrared spectroscopy", Proceedings of SPIE, Optical Tomography and Spectroscopy of Tissue IV, Jun. 29, 2001, pp. 470-481, vol. 4250, SPIE, <https://www.spiedigitallibrary.org/conference-proceedings-of-spie/4250/1/Noninvasive-monitoring-of-brain-tissue-temperature-by-near-infrared-spectroscopy/10.1117/12.434506.short?SSO=1>. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, Mailed Mar. 11, 2021, Corresponding to PCT/IB2020/001037, 13 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, mailed Apr. 4, 2023, corresponding to PCT/EP2022/082341, 33 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, mailed Feb. 1, 2022, corresponding to PCT/IB2021/000649, 18 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, mailed Nov. 15, 2021, corresponding to PCT/IB2021/000517, 15 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, mailed Oct. 10, 2022, corresponding to PCT/IB2022/000373, 16 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, mailed Oct. 19, 2022, corresponding to PCT/EP2022/071467, 16 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, mailed Feb. 14, 2023, corresponding to PCT/EP2022/082162, 33 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, Mailed Dec. 11, 2024, Corresponding to PCT/IB2024/000388, 14 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, mailed May 14, 2025, corresponding to PCT/IB2025/000071, 15 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, mailed Jan. 2, 2023, corresponding to PCT/EP2022/074876, 13 pages. cited by applicant

Invitation to Pay Additional Fees and Partial Search Report mailed Feb. 14, 2023 in related International Application No. PCT/EP2022/082341, 18 pages. cited by applicant

Izutsu, M. et al., "Integrated Optical SSB Modulator/Frequency Shifter", IEEE Journal of Quantum Electronics, Nov. 1981, pp. 2225-2227, vol. QE-17, No. 11, IEEE. cited by applicant

Kang, J. W. et al., "Direct observation of glucose fingerprint using in vivo Raman spectroscopy," Science Advances, Jan. 24, 2020, pp. 1-8, American Association for the Advancement of Science. cited by applicant

Karlsson, C. J. et al., "All-fiber multifunction continuous-wave coherent laser radar at 1.55 μm for range, speed, vibration, and wind measurements", Applied Optics, Jul. 20, 2000, pp. 3716-3726, vol. 39, No. 21, Optical Society of America. cited by applicant

Lai, M. et al., "Perfusion Monitoring By Contactless Photoplethysmography Imaging", 2019 IEEE 16th International Symposium on Biomedical Imaging (ISBI 2019), Venice, Italy, Apr. 8-11, 2019, pp. 1778-1782, IEEE. cited by applicant

Lai, N. et al., "CO₂ Capture With Absorbents of Tertiary Amine Functionalized Nano-SiO₂", Frontiers in Chemistry, Feb. 28, 2020, pp. 1-9, vol. 8, Article 146, www.frontiersin.org. cited by applicant

Lapchuk, A. et al., "Investigation of speckle suppression beyond human eye sensitivity by using a passive multimode fiber and a multimode fiber bundle", Applied Optics, Feb. 21, 2020, pp. 6820-6834, vol. 28, No. 5, Optical Society of America. cited by applicant

Liu, X. et al., "Simultaneous measurements of tissue blood flow and oxygenation using a wearable fiber-free optical sensor", Journal of Biomedical Optics, Jan. 29, 2021, pp. 012705-1 through 012705-15, vol. 26, No. 1, SPIE. cited by applicant

Loi, R. et al., "Transfer Printing of AlGaInAs/InP Etched Facet Lasers to Si Substrates", IEEE Photonics Journal, Nov. 11, 2016, 11 pages, vol. 8, No. 6, IEEE. cited by applicant

Lu, H. et al., "Single-trial estimation of the cerebral metabolic rate of oxygen with imaging photoplethysmography and laser speckle contrast imaging", Optics Letters, Mar. 17, 2015, pp. 1193-1196, vol. 40, No. 7, Optical Society of America. cited by applicant

Mehta, D. S. et al., "Laser speckle reduction by multimode optical fiber bundle with combined temporal, spatial, and angular diversity", Applied Optics, Apr. 11, 2012, pp. 1894-1904, vol. 51, No. 12, Optical Society of America. cited by applicant

Merritt, S. et al., "Monitoring temperature non-invasively using broadband Diffuse Optical Spectroscopy", OSA/FIO, 2004, 1 page, Optical Society of America, <https://opg.optica.org/abstract.cfm?URI=FiO-2004-FTuK4>. cited by applicant

Mosso, E. et al., "Cluster speckle structures through multiple apertures forming a closed curve", Optics Communications, 2010, pp. 1285-1290, Elsevier B.V. cited by applicant

Nabeel, P. M. et al., "Local Pulse Wave Velocity: Theory, Methods, Advancements, and Clinical

Applications”, IEEE Reviews in Biomedical Engineering, Jul. 29, 2019, pp. 74-112, vol. 13, IEEE. cited by applicant

Noriki, A. et al., “45-degree curved micro-mirror for vertical optical I/O of silicon photonics chip”, Optics Express, Jul. 1, 2019, pp. 19749-19757, vol. 27, No. 14, Optical Society of America, <https://doi.org/10.1364/OE.27.019749>. cited by applicant

Poulton, C. V. et al., “Frequency-modulated Continuous-wave LIDAR Module in Silicon Photonics”, OFC, 2015, 4 pages, Optical Society of America. cited by applicant

Qiu, J. et al., “Correcting speckle contrast at small speckle size to enhance signal to noise ratio for laser speckle contrast imaging”, Optics Express, Nov. 15, 2013, pp. 28902-28913, vol. 21, No. 23, Optical Society of America. cited by applicant

Redding, B. et al., “Compact spectrometer based on a disordered photonic chip”, Nature Photonics, Jul. 28, 2013, pp. 746-751, vol. 7, Macmillan Publishers Limited. cited by applicant

Redding, B. et al., “Evanescently coupled multimode spiral spectrometer”, Optica, Aug. 25, 2016, pp. 956-962, vol. 3, No. 9, Optical Society of America. cited by applicant

Robinson, M. B., “Interferometric diffuse correlation spectroscopy improves measurements at long source-detector separation and low photon count rate”, Journal of Biomedical Optics, Sep. 30, 2020, pp. 097004-1 through 097004-12, vol. 25, No. 9, SPIE. cited by applicant

Roelkens, G. et al., “Transfer printing for silicon photonics transceivers and interposers”, 2018 IEEE Optical Interconnects Conference, Jun. 4, 2018, pp. 13-14, IEEE. cited by applicant

Ryckeboer, E., “Spectroscopic Detection of Glucose with a Silicon Photonic Integrated Circuit”, Universiteit Gent, Jan. 1, 2014, 263 pages, ISBN 978-90-8578-688-7, http://www.photonics.intec.ugent.be/download/phd_206.pdf. cited by applicant

Schneider, S. et al., “Optical coherence tomography system mass-producible on a silicon photonic chip”, Optics Express, Jan. 20, 2016, pp. 1573-1586, vol. 24, No. 2, Optical Society of America. cited by applicant

Sdobnov, A. Y. et al. “Speckle dynamics under ergodicity breaking”, Journal of Physics D: Applied Physics, Mar. 26, 2018, pp. 1-21, vol. 51, No. 15, IOP Publishing Ltd. cited by applicant

Shimotsu, S. et al., “Single Side-Band Modulation Performance of a LiNbO3 Integrated Modulator Consisting of Four-Phase Modulator Waveguides”, IEEE Photonics Technology Letters, Apr. 2001, pp. 364-366, vol. 13, No. 4, IEEE. cited by applicant

Subramanian, A. Z. et al., “Silicon and silicon nitride photonic circuits for spectroscopic sensing on-a-chip [Invited]”, Photon. Res., Aug. 28, 2015, pp. B47-B59, vol. 3, No. 5, Chinese Laser Press. cited by applicant

Teng, Zhongshuai et al., “In Vivo Pulse Wave Measurement Through a Multimode Fiber Diffuse Speckle Analysis System”, Frontiers in Physics, Jan. 19, 2021, pp. 1-8, vol. 8, Article 613342, www.frontiersin.org. cited by applicant

Timm, U. et al., “Non-Invasive Optical Real-time Measurement of Total Hemoglobin Content”, Procedia Engineering, 2010, pp. 488-491, Elsevier Ltd. cited by applicant

Tran, T-T-K. et al., “Speckle reduction in laser projection displays through angle and wavelength diversity”, Applied Optics, Feb. 16, 2016, pp. 1267-1274, vol. 55, No. 6, Optical Society of America. cited by applicant

Tuchin, V., “Chapter 8: Coherent Effects at the Interaction of Laser Radiation with Tissues and Cell Flows”, Tissue Optics Light Scattering Methods and Instruments for Medical Diagnostics, 3rd Edition, 2015, pp. 359-417, SPIE. cited by applicant

U.S. Advisory Action for U.S. Appl. No. 17/822,419, dated May 2, 2024, 5 pages. cited by applicant

U.S. Advisory Action for U.S. Appl. No. 17/822,419, dated Oct. 10, 2023, 6 pages. cited by applicant

U.S. Appl. No. 18/991,054, filed Dec. 20, 2024. cited by applicant

U.S. Notice of Allowance for U.S. Appl. No. 17/703,920, dated Jul. 31, 2024, 10 pages. cited by applicant

applicant
U.S. Notice of Allowance from U.S. Appl. No. 17/757,130, dated Jan. 18, 2023, 10 pages. cited by applicant
U.S. Notice of Allowance from U.S. Appl. No. 17/822,419, dated Nov. 6, 2024, 10 pages. cited by applicant
U.S. Office Action for U.S. Appl. No. 17/703,920, dated Apr. 5, 2024, 11 pages. cited by applicant
U.S. Office Action for U.S. Appl. No. 17/822,419, dated Feb. 20, 2024, 12 pages. cited by applicant
U.S. Office Action for U.S. Appl. No. 17/822,419, dated Jun. 12, 2024, 12 pages. cited by applicant
U.S. Office Action for U.S. Appl. No. 17/822,419, dated Nov. 3, 2023, 18 pages. cited by applicant
U.S. Office Action from U.S. Appl. No. 17/711,974, dated Oct. 11, 2024, 18 pages. cited by applicant
U.S. Office Action from U.S. Appl. No. 17/822,419, dated Jul. 20, 2023, 21 pages. cited by applicant
U.S. Office Action from U.S. Appl. No. 17/822,419, dated Mar. 10, 2023, 17 pages. cited by applicant
U.S. Office Action from U.S. Appl. No. 17/934,502, dated Aug. 31, 2023, 6 pages. cited by applicant
U.S. Office Action from U.S. Appl. No. 17/934,502, dated Feb. 1, 2024, 6 pages. cited by applicant
U.S. Office Action from U.S. Appl. No. 17/934,502, dated Jun. 5, 2024, 7 pages. cited by applicant
U.S. Office Action from U.S. Appl. No. 18/778,893, dated Apr. 22, 2025, 9 pages. cited by applicant
U.S. Office Action from U.S. Appl. No. 18/991,054, dated Mar. 3, 2025, 19 pages. cited by applicant
Valley, G.C. et al., “Multimode waveguide speckle patterns for compressive sensing”, Optics Letters, May 23, 2016, pp. 2529-2532, vol. 41, No. 11, Optical Society of America. cited by applicant
International Search Report and Written Opinion of the International Searching Authority, mailed Jun. 23, 2025, corresponding to PCT/IB2025/000070, 15 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) The present application claims priority to and the benefit of U.S. Provisional Application No. 63/726,157, filed Nov. 27, 2024, entitled “WEARABLE DEVICE WITH LIGHT SOURCE AND OPTICAL SENSOR”, the entire content of which is incorporated herein by reference.

FIELD

(1) One or more aspects of embodiments according to the present disclosure relate to biomarker monitoring, and more particularly to a system and method for measuring one or more biomarkers using one or more light sources and one or more optical sensors.

BACKGROUND

(2) Various cardiovascular biomarkers may be clinically significant, including, for example, the blood pressure of a subject.

(3) It is with respect to this general technical environment that aspects of the present disclosure are related.

SUMMARY

(4) According to an embodiment of the present disclosure, there is provided a system, including: a wearable device for being worn by a subject and for measuring a first biomarker of a blood vessel of the subject, the wearable device including: a first light source, and a plurality of optical detectors, the system being configured: to perform a photoplethysmography measurement with each of the plurality of optical detectors, the first biomarker being a blood volume; or to perform a speckleplethysmography measurement with each of the plurality of optical detectors, the first biomarker being a blood flow velocity.

(5) In some embodiments, the system is configured to read interleaved data from the plurality of optical detectors.

(6) In some embodiments, the system is further configured: to read first interleaved data from a first subset of the plurality of optical detectors; and to read second interleaved data from a second subset of the plurality of optical detectors.

(7) In some embodiments, the system is configured: to perform a first measurement, with a first optical detector of the plurality of optical detectors; to perform a second measurement, with a second optical detector of the plurality of optical detectors, the second optical detector being azimuthally separated from the first optical detector; and to determine that a measure of signal quality of the first measurement exceeds a measure of signal quality of the second measurement.

(8) In some embodiments, the system is further configured: in response to the determining that the measure of signal quality of the first measurement exceeds the measure of signal quality of the second measurement, to perform a third measurement, with the first optical detector.

(9) In some embodiments: the system is further configured: to read first interleaved data from a first subset of the plurality of optical detectors; and to read second interleaved data from a second subset of the plurality of optical detectors, wherein the first optical detector is an optical detector of the first subset, and the second optical detector is an optical detector of the second subset.

(10) In some embodiments, the system is further configured: in response to the determining that the measure of signal quality of the first measurement exceeds the measure of signal quality of the second measurement, to prompt a user of the system to perform an azimuthal adjustment of the wearable device.

(11) In some embodiments: the first biomarker is a blood flow velocity, and the plurality of optical detectors includes a first image sensor and a second image sensor azimuthally separated from the first image sensor.

(12) In some embodiments, the first image sensor is a zero chief ray angle image sensor.

(13) In some embodiments, an optical path between the blood vessel of the subject and the first image sensor includes no lenses.

(14) In some embodiments, the wearable device further includes a receiving window, and the length of an optical path between the receiving window and an optical detector of the plurality of optical detectors is less than 15 mm.

(15) In some embodiments, the wearable device further includes a receiving window, and a distance between the receiving window and an optical detector of the plurality of optical detectors is less than 5 mm.

(16) In some embodiments: the wearable device further includes a second light source; and the system is further configured: to perform a photoplethysmography measurement with the second light source and an optical detector of the plurality of optical detectors; or to perform a speckleplethysmography measurement with the second light source and an optical detector of the plurality of optical detectors.

(17) According to an embodiment of the present disclosure, there is provided a system, including: a wearable device for being worn by a subject and for measuring a first biomarker of a blood vessel of the subject, the wearable device including: a plurality of light sources, and a first optical detector, the system being configured: to perform a photoplethysmography measurement with each

of the plurality of light sources, the first biomarker being a blood volume; or to perform a speckleplethysmography measurement with each of the plurality of light sources, the first biomarker being a blood flow velocity.

(18) In some embodiments, the system is configured: to perform a first measurement, with a first light source of the plurality of light sources; to perform a second measurement, with a second light source of the plurality of light sources, the second light source being azimuthally separated from the first light source; and to determine that a measure of signal quality of the first measurement exceeds a measure of signal quality of the second measurement.

(19) In some embodiments, the system is further configured: in response to the determining that the measure of signal quality of the first measurement exceeds the measure of signal quality of the second measurement, to perform a third measurement, with the first light source.

(20) In some embodiments, the system is further configured: in response to the determining that the measure of signal quality of the first measurement exceeds the measure of signal quality of the second measurement, to prompt a user of the system to perform an azimuthal adjustment of the wearable device.

(21) In some embodiments: the wearable device further includes a second optical detector; and the system is further configured: to perform a photoplethysmography measurement with a light source of the plurality of light sources and the second optical detector; or to perform a speckleplethysmography measurement with a light source of the plurality of light sources and the second optical detector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features and advantages of the present disclosure will be appreciated and understood with reference to the specification, claims, and appended drawings wherein:

(2) FIG. 1A is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(3) FIG. 1B is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(4) FIG. 1C is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(5) FIG. 1D is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(6) FIG. 1E is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(7) FIG. 1F is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(8) FIG. 1G is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(9) FIG. 1H is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(10) FIG. 1I is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(11) FIG. 1J is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(12) FIG. 1K is a schematic drawing of a health sensor, according to an embodiment of the present disclosure;

(13) FIG. 2A is a schematic drawing of an images sensor and processor interface, according to an

embodiment of the present disclosure;

(14) FIG. 2B is a schematic drawing of a plurality of images sensors and processor interfaces, according to an embodiment of the present disclosure;

(15) FIG. 2C is a schematic drawing of a plurality of images sensors and processor interfaces, according to an embodiment of the present disclosure;

(16) FIG. 2D is a schematic drawing of a plurality of images sensors and processor interfaces, according to an embodiment of the present disclosure;

(17) FIG. 3A is a schematic drawing of a sensor with four image sensors, over an artery, according to an embodiment of the present disclosure;

(18) FIG. 3B is a graph of data corresponding to the configuration of FIG. 3A, according to an embodiment of the present disclosure;

(19) FIG. 3C is a schematic drawing of a sensor with four image sensors, over an artery, according to an embodiment of the present disclosure; and

(20) FIG. 3D is a graph of data corresponding to the configuration of FIG. 3C, according to an embodiment of the present disclosure; and

(21) FIG. 4 is a flow chart, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(22) The detailed description set forth below in connection with the appended drawings is intended as a description of exemplary embodiments of a wearable device with a light source and an optical sensor provided in accordance with the present disclosure and is not intended to represent the only forms in which the present disclosure may be constructed or utilized. The description sets forth the features of the present disclosure in connection with the illustrated embodiments. It is to be understood, however, that the same or equivalent functions and structures may be accomplished by different embodiments that are also intended to be encompassed within the scope of the disclosure. As denoted elsewhere herein, like element numbers are intended to indicate like elements or features.

(23) Referring to FIG. 1A, a health sensor (e.g., a cuffless blood pressure sensor) may include one or more light sources **110** for illuminating one or more blood vessels in a patient, or “subject”, and optical detectors **120** (e.g., a photodiode (PD) **102** or an image sensor **100** (e.g., a CMOS image sensor (CIS)) for receiving light scattered from the blood vessel (e.g., from blood inside the blood vessel). The health sensor may include (i) a photoplethysmography (PPG) sensor including a source of incoherent light **105** (e.g., S1, in FIG. 1A) and a photodetector (e.g., a single-pixel photodetector such as a photodiode **102** (PD)) for measuring the instantaneous blood volume in the blood vessel, and (ii) a speckleplethysmography (SPG) sensor (including a source of coherent light such as a laser **115** (e.g., S2, in FIG. 1A) and an image sensor **100** (e.g., a CMOS image sensor (CIS) or a charge-coupled device (CCD)) for measuring the instantaneous blood velocity in the blood vessel. Each of the instantaneous blood volume in the blood vessel, and the instantaneous blood velocity may vary with the cardiac cycle; blood pressure (both the systolic blood pressure and the diastolic blood pressure) may be inferred from a time history (or “time series”) of instantaneous blood volume measurements and a time history of instantaneous blood velocity measurements.

(24) FIG. 1B shows the performing of a photoplethysmography measurement (with the blood vessel illuminated by the source of incoherent light, and the photodiode **102** (PD)) being used to detect the light scattering from the blood vessel into the photodiode **102**. FIG. 1C shows the performing of a speckleplethysmography measurement (with the blood vessel illuminated by the source of coherent light S2, and the image sensor **100** (labeled CIS) being used to detect the light scattering from the blood vessel into the image sensor **100**. When a speckleplethysmography measurement is performed, the measured speckle contrast (which is a measurement that is time-averaged over the exposure time of the image sensor **100**) may decrease with increasing blood flow velocity (which may cause the speckle pattern on the image sensor **100** to change more rapidly); as

such, the measured speckle contrast may be used as an indication of blood flow velocity.

(25) Each image (or frame) generated by the image sensor **100** may be processed to calculate a data point of a time series representing a biomarker. One such biomarker may be, as mentioned above, a value that is representative of flow of blood in an artery. In such a case the alignment of the light source **110** and image sensor **100** may be such that the light scattered by the flowing blood in the artery is able to be captured by the imaging device.

(26) For the case in which a health sensor based on laser speckle imaging (e.g., a health sensor based on speckleplethysmography) is used in a wearable device to measure the blood flow in a particular anatomical feature such as an artery, for example the radial artery at the wrist, a need to align the sensor with this anatomical feature may make the sensor performance highly dependent on device placement. The small radial artery diameter (2-3 mm) combined with the limited sensing volume offered by a sensor consisting of a single light source **110** and single image sensor **100** may contribute to this dependency on device placement.

(27) Placing a wearable device over a non-visible blood vessel such as the radial artery on the volar side of the wrist may be challenging, especially when the body of the wearable device itself obstructs the view of skin when placed in the general area of the artery. This limits the user's ability to align the emission and detection windows of the wearable device with a location on the skin that might have been identified pre-placement via ultrasound imaging or palpitation. This last issue may be mitigated somewhat using datums on the body of the wearable device that may be positioned relative to landmarks of the subject's anatomy for alignment to the wearable device's optimal sensing point. A multi-step placement process involving first locating the blood vessel (e.g., by palpitation) relative to landmarks of the subject and then placing the wearable device at a location defined by the landmarks may be challenging for an untrained user, however. As used herein, the subject is a person wearing the wearable device, and the "user" is a person operating the wearable device. The user may be the subject or another person, e.g., a clinician.

(28) Various aspects of the design of the wearable device may affect sensitivity to alignment. FIG. **1D** shows an illumination shape in which the laser beam **125** is highly elliptical, so as to illuminate more tissue while not greatly impacting source-detector separation. Such a configuration may impact signal-to-noise ratio (SNR) performance by the addition of light not travelling through the radial artery, however. FIG. **1E** shows a configuration with multiple emitters (e.g., multiple lasers **115**) illuminating, with multiple respective laser beams **125**, the tissue of the subject, and the artery. In such an embodiment, the lasers **115** may be turned on one at a time, to identify the laser **115** associated with the best signal-to-noise ratio performance. FIG. **1F** shows an embodiment with multiple optical detectors **120**. For example, three image sensors **100** may be operated in "multi-sensor" mode where frames are captured sequentially from each sensor. The effective frame rate may be divided by 3 (in the case of using 3 image sensors **100**). In some such embodiments, the image sensor **100** with the best performance (which may depend on the location, relative to the light source **110** and the artery, of each of the image sensors **100**) may be identified and the wearable device may then transition to using only this image sensor **100**, at full frame rate. FIGS. **1G-1I** show embodiments with multiple light sources **110** (e.g., multiple lasers **115** illuminating the tissue of the subject with multiple respective laser beams **125**) and multiple optical detectors **120**. FIGS. **1J** and **1K** show, respectively, an embodiment with multiple optical detectors **120** and one light source **110**, and with multiple light sources **110** and one optical detector **120**.

(29) As mentioned above, to reduce the positional sensitivity of a sensor the number of transmitting elements or the number of receiving elements may be increased. In the case of a laser-based speckle imaging sensor (e.g., in the case of a speckleplethysmography sensor) this may be accomplished by increasing the number of lasers **115** or the number of image sensors **100** (or both). If multiple lasers **115** (that are not phase coherent with each other) are used, then only one laser **115** may be turned on at a time, to avoid the reduction in speckle contrast that may otherwise occur when light from several lasers **115** is received at an image sensor **100**. For a convex, generally

cylindrical part of the subject (such as a finger, a limb, the neck, or the torso of the subject) (which may be referred to as a “convexity” of the subject) with a blood vessel (e.g., the radial artery) extending generally parallel to, and offset from a central axis of, the part (as, for example, the radial artery at the wrist extends generally parallel to the arm and near the skin) the degree of freedom in which accurate placement may be important may be the “azimuthal” position of the sensor (which for a band securing the wearable device to the wrist may be adjusted by rotating the band (along with the wearable device) around the wrist (about the longitudinal axis of the forearm)).

(30) Using multiple image sensors **100** offers the capability to detect over a broader physical space and cover regions where the scattered light would be generally lost when a single sensor is employed. FIG. 2A shows a simplified signal chain of the receiving elements of the wearable device. An image sensor **100** transmits image data (frames) to a processor **205** (e.g., a processing circuit) (such as a microcontroller (MCU) or field programmable gate array (FPGA)). The processor **205** receives frame data from the image sensor **100** via an interface **210** (such as a parallel bus interface or a serial interface such as the Camera Serial Interface (CSI) (promulgated by the Mobile Industry Processor Interface (MIPI) Alliance). The frames may be queued before frame processing. The queue may have a length of zero in which case frame processing occurs immediately. Frame processing extracts information from each frame received, resulting in value representative of the state of the speckle in the imaged frame captured at a time t . This value may be, for example, a measured speckle contrast. The image sensor **100** operates at a frame rate which may define the sample rate of the time series.

(31) In some embodiments, multiple image sensors **100** are operated in parallel as illustrated in FIG. 2B. In the example shown, four image sensors **100** are shown but the design may be employed with more or fewer image sensors **100**. The processor **205** in this embodiment need not be a single processor **205** but may be a distributed processor **205**, including a plurality of processors **205**.

(32) In some embodiments, multiple image sensors **100** share a communication bus and an interface **210** with the processor **205**, as shown in FIG. 2C. The processor **205** may have a limited number of sensor interfaces **210**. An aggregator or switch or multiplexer (external to the processor **205**) may be connected to the image sensors **100**, and, in operation, may interleave their frame streams and transmit the resulting composite stream to the processor **205** through a single CIS interface **210**. In some embodiments, a switch or multiplexer is employed to switch between generated frames, so the processor **205** receives an interleaved sequence. A switch may also be employed in a non-interleaved or non-simultaneous sampling embodiment. In such an embodiment, each image sensor **100** may be monitored for a few seconds sequentially to assess signal quality. The optimal image sensor **100** may then be identified. In this embodiment the image sensors **100** may capture frames at a simultaneous point in time at a frame rate F and transmit those frames to the processor **205** at rate NF (where N is the number of image sensors **100**; e.g., $N=4$ in the embodiment of FIG. 2C). In some embodiments, the image sensors **100** instead capture frames sequentially, so that the rate at which each sensor runs is F/N , and the rate at which the frames are sent to the processor **205** is equal to F . In both cases the captured frames may be de-interleaved, e.g., in the processor **205** (or in a separate de-interleaving circuit) to separate into N respective streams the frames from the N image sensors **100**. The processor **205** may then process the frames provided by the image sensors **100** in parallel, as illustrated in FIG. 2C.

(33) In some embodiments, e.g., if the processor **205** lacks sufficient processing resources to process the frames in parallel, the frames are processed one at a time (e.g., in series) in the processor **205** (instead of being processed in parallel). In such an implementation, an example of which is shown in FIG. 2D, the frames in the stream provided by the image sensors **100** are processed sequentially and the corresponding data points are de-interleaved to produce a corresponding respective time series for each of the image sensors **100**.

(34) In some embodiments, the image sensors **100** are grouped into subsets, and one subset of image sensors **100** is read out at a time, in parallel or in an interleaved manner. For example, after one or more frames have been read from each of a first subset of image sensors **100**, the system may switch to reading out a second subset of the image sensors **100**, e.g., in an interleaved manner. Once one or more frames have been read from each of the second subset of image sensors **100**, the system may switch to a third subset of the image sensors **100**, and so on, or it may switch back to the first subset of the image sensors **100**.

(35) FIG. 3A shows (i) a speckleplethysmography sensor including four image sensors **100** and a single light source **110**, and (ii) an artery that is the target of the speckleplethysmography sensor. The resulting representative time series from CIS **1** through **4** are shown in plot A through D respectively, of FIG. 3B; plot E of FIG. 3B shows the composite time series before de-interleaving.

(36) FIG. 3C shows the speckleplethysmography sensor of FIG. 3A after the position of the wearable device with respect to the artery has changed (e.g., as a result of a user's moving of the wearable device). The effect on the resulting time series may be seen in the plots in FIG. 3D (in which representative time series from CIS **1** through **4** are shown in plot A through D respectively, and plot E shows the composite time series before de-interleaving); the magnitude of the time series data collected by CIS **2** and **4** is increased in comparison with the time series data corresponding to the position of FIG. 3A.

(37) In some embodiments, a speckleplethysmography sensor utilizing multiple image sensors **100** may at startup, or periodically, be operated with all of the image sensors **100** and may then transition to a mode in which only a single image sensor **100** operates, e.g., to conserve power, or to permit the capturing and processing of images at a higher rate. The single image sensor **100** used after the transition may be the image sensor **100** generating the highest signal quality before the transition, where signal quality may be measured by a suitable algorithm (e.g. maximum peak-to-peak amplitude). FIG. 4 shows a flow chart corresponding to such a method. In the method of FIG. 4, frames are captured simultaneously from multiple image sensors **100**, at **405**; the frames are processed, at **410**, to create time series data corresponding to each image sensor **100**; the signal quality for each such time series is calculated, at **415**; the image sensor **100** producing signals with the highest signal quality is identified, at **420**, and the system switches, at **425**, to a mode in which it uses only this identified image sensor **100**. A transition back to a mode in which all of the image sensors **100** are used may be triggered by an external stimulus (e.g., an accelerometer reading exceeding a threshold) or by a real-time evaluation of the sensor time series data (e.g., peak-to-peak speckle contrast amplitude being below a threshold).

(38) In some embodiments, a difference in signal quality between two image sensors **100** may cause the wearable device to prompt a user to perform an azimuthal adjustment of the wearable device, so as to improve the positioning of the wearable device. For example, in the situation illustrated in FIG. 3A, the system may infer, from the higher signal quality at CIS **1**, that an azimuthal displacement of the wearable device, so that the position of the artery is nearer the center of the constellation of image sensors **100**, may improve the signal quality at one or more of the image sensors **100**, and the wearable device may prompt the user accordingly.

(39) In some embodiments, multiple photodiodes **102**, or multiple light sources **110** (e.g., multiple coherent light sources or multiple incoherent light sources) may be used in an analogous manner, to reduce the sensitivity to placement of the wearable device, or to generate information based on which a user may be prompted to adjust the azimuthal position of the wearable device.

(40) In some embodiments with multiple image sensors **100**, each image sensor **100** may receive light through a respective separate window in the wearable device (which may be shared with a respective photodiode **102**), and a separate window may be used for each light source **110**, or for all of the light sources **110**.

(41) An image sensor **100** designed for imaging applications may include a micro-lens array that includes one micro-lens for each pixel in the image sensor **100**. Each micro-lens may concentrate

the received light on the light-sensitive portion of the corresponding pixel (in some devices part of the area of the pixels is allocated to a control and readout circuit and is not light-sensitive). Such a microlens may be offset from the center of the light-sensitive portion of the corresponding pixel if the chief ray angle for the pixel, in an optical system for which the image sensor **100** is intended, is not zero, i.e., if the chief ray corresponding to the pixel is not perpendicular to the plane of the image sensor **100**. This may be because the pixel is not near the center of the array (e.g., because the pixel is near the edge of the array). The offsetting of microlenses from the light-sensitive portion of the corresponding pixels may result in vignetting (lower exposure at and near the edges of the image sensor **100**) in applications (such as the optodes described herein) in which the illumination is generally diffuse Lambertian illumination, which is approximately isotropic over a large solid angle, because in such a configuration a greater proportion of the received light may miss the light-sensitive portion of the corresponding pixel if the microlens is offset from the light-sensitive portion of the corresponding pixel. As such, in some embodiments, an image sensor **100** that does not include a micro-lens array, or an image sensor **100** that includes a micro-lens array and that is a zero chief ray angle (CRA) image sensor (e.g., an image sensor **100** in which each micro-lens is aligned with (not offset from) the light-sensitive portion of the corresponding pixel) may be used. As used herein, a zero chief ray angle sensor is a sensor (i) that does not include a microlens array or (ii) that includes a micro-lens array in which each chief ray angle is small (e.g., less than 5 degrees) (where the chief ray is a ray that passes through the center of the aperture and ends at the center of the light-sensitive area of the pixel, and the chief ray angle is the angle between the chief ray and a line perpendicular to the plane of the detector **120**).

(42) The distance from the sample (e.g., from the skin of the subject) to the image sensor **100** may be less than 5 mm (in a straight line) or less than 15 mm along the optical path (the latter distance being greater than the former if, for example, a folding mirror is used to cause the light from the sample to propagate parallel to the surface of the sample for some distance before reaching the image sensor **100**).

(43) The wearable device may be coupled to various other devices and systems which may perform functions of data analysis, data storage, and user interaction. For example, the wearable device may be connected (e.g., via a wireless connection) to a mobile device (such as a mobile telephone or portable (e.g., laptop) computer) and a portion of the data analysis used to convert the signals received from the optical sensors to one or more biomarkers (e.g., blood flow velocity, blood volume, or blood pressure) may be performed in the mobile device. The mobile device may also be connected to the cloud (e.g., to one or more servers connected to the Internet) and data processing and storage may be performed on the cloud. As another example, if an adjustment is to be made (e.g., an adjustment to the azimuthal position of the wearable device), the wearable device may cause the mobile device to give the user a suitable corresponding prompt.

(44) As used herein, “a portion of” something means “at least some of” the thing, and as such may mean less than all of, or all of, the thing. As such, “a portion of” a thing includes the entire thing as a special case, i.e., the entire thing is an example of a portion of the thing. As used herein, when a second quantity is “within Y” of a first quantity X, it means that the second quantity is at least X-Y and the second quantity is at most X+Y. As used herein, when a second number is “within Y %” of a first number, it means that the second number is at least $(1-Y/100)$ times the first number and the second number is at most $(1+Y/100)$ times the first number. As used herein, the word “or” is inclusive, so that, for example, “A or B” means any one of (i) A, (ii) B, and (iii) A and B.

(45) Each of the terms “processing circuit” and “means for processing” is used herein to mean any combination of hardware, firmware, and software, employed to process data or digital signals. Processing circuit hardware may include, for example, application specific integrated circuits (ASICs), general purpose or special purpose central processing units (CPUs), digital signal processors (DSPs), graphics processing units (GPUs), and programmable logic devices such as field programmable gate arrays (FPGAs). In a processing circuit, as used herein, each function is

performed either by hardware configured, i.e., hard-wired, to perform that function, or by more general-purpose hardware, such as a CPU, configured to execute instructions stored in a non-transitory storage medium. A processing circuit may be fabricated on a single printed circuit board (PCB) or distributed over several interconnected PCBs. A processing circuit may contain other processing circuits; for example, a processing circuit may include two processing circuits, an FPGA and a CPU, interconnected on a PCB.

(46) As used herein, when a method (e.g., an adjustment) or a first quantity (e.g., a first variable) is referred to as being “based on” a second quantity (e.g., a second variable) it means that the second quantity is an input to the method or influences the first quantity, e.g., the second quantity may be an input (e.g., the only input, or one of several inputs) to a function that calculates the first quantity, or the first quantity may be equal to the second quantity, or the first quantity may be the same as (e.g., stored at the same location or locations in memory as) the second quantity.

(47) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the inventive concept. As used herein, the terms “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent deviations in measured or calculated values that would be recognized by those of ordinary skill in the art.

(48) Any numerical range recited herein is intended to include all sub-ranges of the same numerical precision subsumed within the recited range. For example, a range of “1.0 to 10.0” or “between 1.0 and 10.0” is intended to include all subranges between (and including) the recited minimum value of 1.0 and the recited maximum value of 10.0, that is, having a minimum value equal to or greater than 1.0 and a maximum value equal to or less than 10.0, such as, for example, 2.4 to 7.6.

Similarly, a range described as “within 35% of 10” is intended to include all subranges between (and including) the recited minimum value of 6.5 (i.e., $(1-35/100)$ times 10) and the recited maximum value of 13.5 (i.e., $(1+35/100)$ times 10), that is, having a minimum value equal to or greater than 6.5 and a maximum value equal to or less than 13.5, such as, for example, 7.4 to 10.6. Any maximum numerical limitation recited herein is intended to include all lower numerical limitations subsumed therein and any minimum numerical limitation recited in this specification is intended to include all higher numerical limitations subsumed therein.

(49) It will be understood that when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present. As used herein, “generally connected” means connected by an electrical path that may contain arbitrary intervening elements, including intervening elements the presence of which qualitatively changes the behavior of the circuit. As used herein, “connected” means (i) “directly connected” or (ii) connected with intervening elements, the intervening elements being ones (e.g., low-value resistors or inductors, or short sections of transmission line) that do not qualitatively affect the behavior of the circuit.

(50) Although exemplary embodiments of a wearable device with a light source and an optical sensor have been specifically described and illustrated herein, many modifications and variations will be apparent to those skilled in the art. Accordingly, it is to be understood that a wearable device with a light source and an optical sensor constructed according to principles of this disclosure may be embodied other than as specifically described herein. The invention is also defined in the following claims, and equivalents thereof.

Claims

1. A system, comprising: a wearable device for being worn by a subject and for measuring a first biomarker of a blood vessel of the subject, the wearable device comprising: a first light source, and a plurality of optical detectors, the system being configured: to perform a photoplethysmography measurement with each of the plurality of optical detectors, the first biomarker being a blood volume; or to perform a speckleplethysmography measurement with each of the plurality of optical

detectors, the first biomarker being a blood flow velocity, wherein the wearable device further comprises a receiving window, and the length of an optical path between the receiving window and an optical detector of the plurality of optical detectors is less than 15 mm.

2. The system of claim 1, wherein the system is configured to read interleaved data from the plurality of optical detectors.

3. The system of claim 2, wherein the system is further configured: to read first interleaved data from a first subset of the plurality of optical detectors; and to read second interleaved data from a second subset of the plurality of optical detectors.

4. The system of claim 1, wherein the system is configured: to perform a first measurement, with a first optical detector of the plurality of optical detectors; to perform a second measurement, with a second optical detector of the plurality of optical detectors, the second optical detector being azimuthally separated from the first optical detector; and to determine that a measure of signal quality of the first measurement exceeds a measure of signal quality of the second measurement.

5. The system of claim 4, wherein the system is further configured: in response to the determining that the measure of signal quality of the first measurement exceeds the measure of signal quality of the second measurement, to perform a third measurement, with the first optical detector.

6. The system of claim 5, wherein: the system is further configured: to read first interleaved data from a first subset of the plurality of optical detectors; and to read second interleaved data from a second subset of the plurality of optical detectors, wherein the first optical detector is an optical detector of the first subset, and the second optical detector is an optical detector of the second subset.

7. The system of claim 1, wherein: the first biomarker is a blood flow velocity, and the plurality of optical detectors comprises a first image sensor and a second image sensor azimuthally separated from the first image sensor.

8. The system of claim 7, wherein the first image sensor is a zero chief ray angle image sensor.

9. The system of claim 8, wherein an optical path between the blood vessel of the subject and the first image sensor includes no lenses.

10. The system of claim 1, wherein the distance between the receiving window and the optical detector is less than 5 mm.

11. The system of claim 1, wherein: the wearable device further comprises a second light source; and the system is further configured: to perform a photoplethysmography measurement with the second light source and an optical detector of the plurality of optical detectors; or to perform a speckleplethysmography measurement with the second light source and an optical detector of the plurality of optical detectors.

12. A system, comprising: a wearable device for being worn by a subject and for measuring a first biomarker of a blood vessel of the subject, the wearable device comprising: a first light source, and a plurality of optical detectors, the system being configured: to perform a photoplethysmography measurement with each of the plurality of optical detectors, the first biomarker being a blood volume; or to perform a speckleplethysmography measurement with each of the plurality of optical detectors, the first biomarker being a blood flow velocity wherein the system is configured: to perform a first measurement, with a first optical detector of the plurality of optical detectors; to perform a second measurement, with a second optical detector of the plurality of optical detectors, the second optical detector being azimuthally separated from the first optical detector; to determine that a measure of signal quality of the first measurement exceeds a measure of signal quality of the second measurement; in response to the determining that the measure of signal quality of the first measurement exceeds the measure of signal quality of the second measurement, to perform a third measurement, with the first optical detector; and in response to the determining that the measure of signal quality of the first measurement exceeds the measure of signal quality of the second measurement, to prompt a user of the system to perform an azimuthal adjustment of the wearable device.

13. A system, comprising: a wearable device for being worn by a subject and for measuring a first biomarker of a blood vessel of the subject, the wearable device comprising: a plurality of light sources, and a first optical detector, the system being configured: to perform a photoplethysmography measurement with each of the plurality of light sources, the first biomarker being a blood volume; or to perform a speckleplethysmography measurement with each of the plurality of light sources, the first biomarker being a blood flow velocity, wherein the system is configured: to perform a first measurement, with a first light source of the plurality of light sources; to perform a second measurement, with a second light source of the plurality of light sources, the second light source being azimuthally separated from the first light source; to determine that a measure of signal quality of the first measurement exceeds a measure of signal quality of the second measurement; in response to the determining that the measure of signal quality of the first measurement exceeds the measure of signal quality of the second measurement, to perform a third measurement, with the first light source; and in response to the determining that the measure of signal quality of the first measurement exceeds the measure of signal quality of the second measurement, to prompt a user of the system to perform an azimuthal adjustment of the wearable device.

14. The system of claim 13, wherein: the wearable device further comprises a second optical detector; and the system is further configured: to perform a photoplethysmography measurement with a light source of the plurality of light sources and the second optical detector; or to perform a speckleplethysmography measurement with a light source of the plurality of light sources and the second optical detector.
